

PARTS LIST

Semiconductors:

IC1	- 7805 voltage regulator
IC2	- PIC16F877A microcontroller
IC3	- LM35 temperature sensor
LED1	- 5mm light-emitting diode

Resistors (all 1/4-watt, 25% carbon):

R1	- 10-kilo-ohm
R2	- 470-ohm
VR1	- 10-kilo-ohm preset

Capacitors:

C1	- 1000µF, 25V electrolytic
C2	- 0.1µF ceramic
C3-C4	- 22pF ceramic

Miscellaneous:

X _{TAL1}	- 4MHz crystal oscillator
D1-D4	- IN4007 diode
S1	- Push-to-on switch
X1	- 230V AC primary to 0-9V, 300mA secondary transformer
	- HD44780 based 16x2 LCD display

The sensor can measure temperature in the range of 0 to 100°C, i.e., the output of the sensor varies from 0 to 1000 mV. The LM35 operates over the temperature range of -55° to +150°C, while the LM35C is rated for a -40°C to +110°C range (-10°C with improved accuracy).

Pin layout of the sensor is as follows:

Pin 1-V_{DD}

Pin 2-Output of the sensor

Pin 3-V_{SS}

PIC16F877A microcontroller. IC PIC16F877A is an 8-bit microcontroller with 8kx14-bit flash program memory, 368 bytes of RAM and many other extra peripherals like ADC, universal synchronous asynchronous receiver transmitter, master synchronous serial port, timers, compare capture and pulse-width modulation modules, and analogue comparators. It is based on the reduced instruction set computer (RISC) architecture.

The microcontroller processes the sensor output to compute the temperature in degree Celsius. The internal ADC of the microcontroller is used to convert the analogue output of the sensor into its equivalent digital value.

The internal ADC of the microcontroller has eight channels of analogue

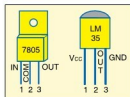


Fig. 3: Pin configurations of 7805 and LM35

input and gives 10-bit digital output. In this project, the reference voltage to the ADC is the same as the supply voltage to the microcontroller, i.e., 5V. The resolution of the ADC can be calculated as follows:

$$\text{Resolution} = \frac{V_{\text{ref}}}{(1024-1)}$$

(as it is a 10-bit ADC)

$$= 5/1023$$

$$= 4.887 \text{ mV}$$

It means that for 4.887mV change in the analogue input, the ADC output changes by binary '1' with a reference voltage of 5V.

Analogue output of the sensor at its pin 2 is connected to Port A at RA0 for conversion into digital equivalent.

The LCD. A 16x2 LCD based on HD44780 controller is used for displaying the temperature. The control lines EN, R/W and RS of the LCD module are connected to pins RA1, RA2 and RA3 of Port A of the microcontroller, respectively. The commands and the data to be displayed are sent to the LCD module in the nibble mode from Port D of the microcontroller. The higher four bits of the LCD (D4 through D7) are connected to the lower nibble of Port D (RD0 through RD3).

Software description

The software code is written in 'C' language and compiled using Hitech C cross compiler in MPLAB IDE. The software performs the following operations in an infinite loop:

1. Initiate analogue-to-digital conversion and obtain the result
2. Calculate the equivalent voltage value from the ADC result
3. Calculate the temperature in degree Celsius from the voltage value
4. Display the temperature on the LCD

The voltage output (in volts) of the sensor is:

$$\text{ADC result} \times 5$$

$$1023$$

The temperature in degree Celsius

is:

$$\frac{\text{Sensor output} \times 1000}{10}$$

$$= \text{Sensor output} \times 100$$

The Hitech C cross-compiler provides floating-point library support that is required for doing the above calculations.

MPLAB IDE and Hitech C compiler. MPLAB IDE is a very powerful software development tool for Microchip products (microcontrollers). It is available for download at 'www.microchip.com' free of cost. It consists of tools like text editor, cross-assembler, cross-compiler and simulator.

Hitech C cross compiler is meant for Microchip PIC10/12/16 series of microcontrollers. Its Lite edition comes free with newer versions of MPLAB IDE like MPLAB v8.2 and v8.3. It can also be downloaded for free from www.htsoft.com.

Compiling the program using MPLAB IDE. The steps follow:

1. Create project file and add source files. In the menu bar, click Project → Project Wizard. The project wizard dialogue box appears. Click 'Next.' In 'Next' window, select the device as PIC16F877A from the drop-down menu. Click 'Next' and select 'Hitech Universal Tool' suite from the drop-down menu. Click 'Next,' name your project file as 'tempr' and specify its location. The file is automatically saved with '.mcp' extension. Click 'Next' and add source files tempr.c, lcd.c and delay.c to your project. If you want to create the source files on your own, you can skip the above step. Click Next → Finish button. Now your project is created and the source files are added to your project.

2. Create and add source files of your own. After creating the project, proceed with typing the code. Open a new text file by clicking 'New' in 'File' menu. Type the code in the text editor and save it with '.c'